



SMS API Developer Guide

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Contents

About this guide	v
Extra reading	v
Standards and conventions	vi
SMS API introduction	1
Before you begin	1
Sending requests to the SMS API	3
Sending mobile terminated SMS messages	4
MT SMS pathway	4
Authentication	4
Request application/x-www-form-urlencoded	5
Schema	5
Creating 16-bit Unicode messages	7
Example	7
Response text/html	8
Example	8
Requesting a status update for existing SMS messages	9
Authentication	9
Request application/x-www-form-urlencoded	9
Schema	9
Example	9
Response application/xml	10
Schema	10
Example	10
Cancelling undelivered SMS messages	11
Authentication	11
Request application/x-www-form-urlencoded	11
Schema	11
Example	11
Response application/xml	12
Schema	12
Example	12

Receiving mobile originated SMS messages	13
Set up a custom HTTP endpoint application	13
MO message format	13
Provide details to Eseye Support	14
Enable reception of MO SMS messages	14
Test MO SMS message reception	15
Delivery receipts for MT and MO SMS messages	16
Authentication	16
Response text/html	16
Example	16
Troubleshooting errors	17
Coding non-alphanumeric ASCII characters	19

About this guide

This guide details how to send and receive SMS messages between your server and a SIM, using the SMS API.

Extra reading

For general information, see the online help: <https://docs.eseye.com/Content/Home.htm>

You can manage SIMs through the [Infinity Classic web interface](#).

To understand SMS API charges, see [Understanding SMS charges on the CSV invoice](#) in the online help.

Standards and conventions

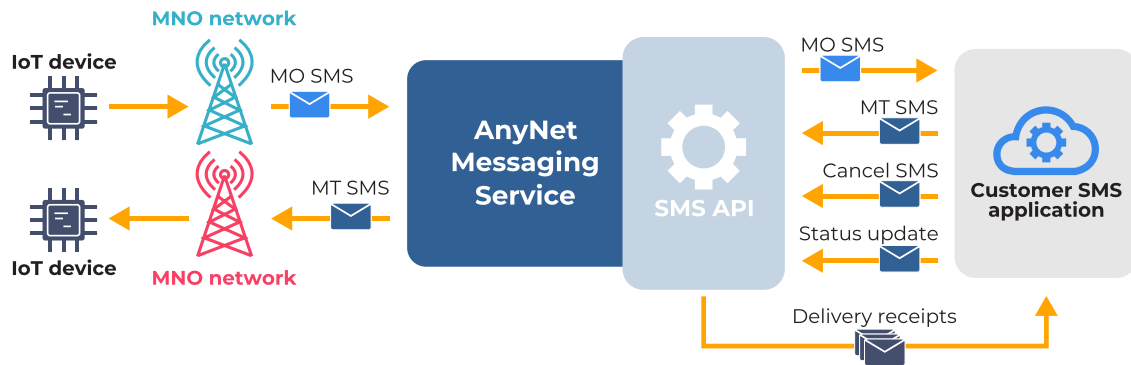
This guide uses consistent visual cues and standard text styles to help you locate and interpret information easily.

Style	Description
 Note	Extra information or a recommendation related to the current topic.
 Tip	Good-to-know information that helps users complete a procedure or understand a topic.
 Warning	Information that alerts the user about significant or critical actions or outcomes.

<i>Title names</i>	Window or section names, denoting a title, appear in italics.
Field or button name	Element names in a user interface, for example fields and buttons, appear in bold.
Ctrl+X; Ctrl+click	A key combination with a plus sign separating two key names or a key name and a mouse action, indicates that you hold down the first key while pressing the second key or performing the mouse action.
Cross reference <i>Title</i> and page	Cross references appear in italics, for example: For more information, see <i>Copyright</i> on page ii. Select the cross reference to view it.
Hyperlinks	Underlined cross references are hyperlinks to electronic forms of the document. Select the hyperlink to open the cross reference.
AT Commands	
Commands	Command formats are displayed in monospaced typeface.
<Parameter>	Angle brackets enclose the AT Command parameter, for example <topic>. The brackets do not appear in the command line.
"ParameterString"	Quotation marks enclose parameter strings.
<i>CommandValue</i>	Italics in a command depicts values or examples that need replacing with your specific parameters.
[CommandOptionalEntry]	Square brackets display optional entries.
ATCommandResponse	Returned responses to AT Commands are displayed in monospaced bold typeface.
<ASCIICHARACTERS>	Returned ASCII characters are in uppercase.

SMS API introduction

The SMS API enables you to send and receive text and short messages to and from SMS-capable devices over internet and cellular networks. The SMS API ensures the message is compatible for network delivery. It also notifies the sender about the message status, so that you can track if the message was delivered or not.



You can send the following requests to the SMS API:

- [MT SMS](#) – sends a mobile terminated (MT) SMS messages to an IoT device.

You send SMS messages to both AnyNet SIMs and third party SIMs. However, SMS messages sent to and received from third party SIMs will incur extra costs. For information about SMS API billing, see [Understanding SMS charges on the CSV invoice](#).

- [Cancel SMS](#) – cancels a previously send MT SMS message.
- [Status update](#) – requests the delivery status of an SMS message.

You can receive the following messages from the SMS API:

- [MO SMS](#) – the AnyNet Messaging Service intercepts and routes mobile originated (MO) SMS messages (sent to a specific API MSISDN) to the customer's SMS application.
- [Delivery receipts](#) – returned to the customer's SMS application or alternatively to a different application.

Before you begin

Eseye will onboard you when you subscribe to SMS API.

You will need:

- A Infinity Classic account, with all your AnyNet SIMs linked to this account.
- SMS Allowlist – contains a list of permissible MSISDNs for mobile originated (MO) and mobile terminated (MT) messages. Messages sent to or received from these MSISDNs are processed.
- SMS Blocklist – contains a list of banned MSISDNs for MO and MT messages. Messages sent to or received from these MSISDNs are blocked.
- Optional server IP address Allowlist – contains a list of permissible IP addresses that can make incoming requests.
- Mobile Originated (MO) URL – where the SMS API will post incoming MO messages.
- Receipt URL – where the SMS API will post delivery receipts.

Eseye will provide you with the following:

- SMS API account username.
- SMS API account password.
- SMS API destination MSISDN that is routed to SMS API.

[Request SMS API](#)



Eseye does not supply historic data. You must store any data you want to retain.

Sending requests to the SMS API

Send SMS API requests to the following URL:

- `https://messaging.eseye.com/<request>/index/post`

where `<request>` is the SMS API request.

Sending mobile terminated SMS messages

Delivers Mobile Terminated (MT) SMS messages to an AnyNet or third party SIM when the following conditions are met:

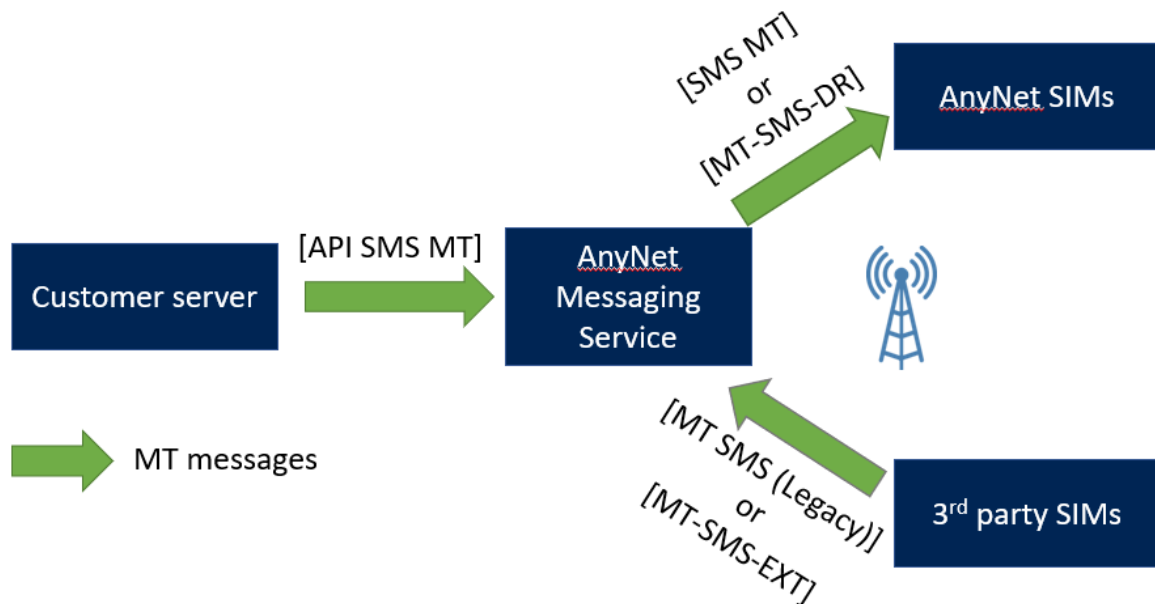
- The authenticated user has access rights to send an SMS
- If an optional IP address Allowlist exists, the source device is listed on it
- The SIM is not barred from receiving SMS messages



Enabling or disabling SMS Messages is linked to the SIM package, which is outside the scope of the SMS API.

MT SMS pathway

The following diagram displays the main pathways used by mobile terminated SMS, with the codes for the pathway in square brackets. The SMS API serves the message to the AnyNet Messaging Service, which then distributes the message to the correct SIM.



Authentication

- SMS API credentials

Request application/x-www-form-urlencoded

Schema

Parameters	Description	Required?
username	The Eseye-assigned SMS API username	✓
password	The Eseye-assigned SMS API password	✓
destination	The destination Mobile Station International ISDN Number (MSISDN).	✓
text	The SMS message content. For 7-bit mode, SMS API only supports characters from the ETSI USSD default alphabet, as defined in GSM 03.38. Uses <code>mode</code> to validate how the data is transported. If both <code>text</code> and <code>short_message</code> are defined, then <code>text</code> is given priority.  Use <code>data_coding</code> and <code>short_message</code> for 16-bit Unicode messages.  For a quick reference on non-alphanumeric ASCII characters, see <i>Coding non-alphanumeric ASCII characters</i> on page 19.	✓ (or <code>short_message</code>)
short_message	The hex-encoded short message octets (user data). Uses <code>data_coding</code> to validate how the data is transported. If both <code>text</code> and <code>short_message</code> are defined, then <code>text</code> is given priority.  Use <code>data_coding</code> and <code>short_message</code> for 16-bit Unicode messages. Encode 16-bit ASCII characters using the Unicode equivalent. For example, where A is 65 (8-bit Hex), instead use 0065 (16-bit Unicode).	✓ (or <code>text</code>)
message_id	A user-generated unique message identifier for tracking responses.	
receipt_url	The URL where the delivery receipt is sent.	
source	The source MSISDN. If blank, then the AnyNet Messaging Service sends the message.	

Parameters	Description	Required?																				
mode	How <code>text</code> data is transported, either as: <table><thead><tr><th>Value</th><th>Description</th><th>Maximum length</th></tr></thead><tbody><tr><td>7</td><td>7-bit encoded (default)</td><td>160 characters</td></tr><tr><td>8</td><td>8-bit encoded</td><td>140 bytes</td></tr></tbody></table>  If the <code>mode</code> value is left blank when <code>text</code> is specified, the default value is used.	Value	Description	Maximum length	7	7-bit encoded (default)	160 characters	8	8-bit encoded	140 bytes												
Value	Description	Maximum length																				
7	7-bit encoded (default)	160 characters																				
8	8-bit encoded	140 bytes																				
expiry	The date and time that the message will expire if undelivered, in the format: YYYY-MM-DD hh:mm:ss																					
data_coding	How the <code>short_message</code> data is processed. For more information, see Data Coding Scheme. Allowed values include: <table><thead><tr><th>Value</th><th>Character/Data set</th><th>Message class</th><th>Maximum length</th></tr></thead><tbody><tr><td>0</td><td>GSM 7-bit, ASCII (default)</td><td>Default</td><td>160 characters</td></tr><tr><td>4</td><td>8-bit data</td><td>Default</td><td>140 bytes</td></tr><tr><td>8</td><td>UCS2, 16-bit</td><td>Default</td><td>70 characters</td></tr><tr><td>F6</td><td>8-bit data</td><td>Class 2 (SIM/USIM-specific)</td><td>140 bytes</td></tr></tbody></table>  If the <code>data_coding</code> value is left blank when <code>short_message</code> is specified, the default value is used.	Value	Character/Data set	Message class	Maximum length	0	GSM 7-bit, ASCII (default)	Default	160 characters	4	8-bit data	Default	140 bytes	8	UCS2, 16-bit	Default	70 characters	F6	8-bit data	Class 2 (SIM/USIM-specific)	140 bytes	
Value	Character/Data set	Message class	Maximum length																			
0	GSM 7-bit, ASCII (default)	Default	160 characters																			
4	8-bit data	Default	140 bytes																			
8	UCS2, 16-bit	Default	70 characters																			
F6	8-bit data	Class 2 (SIM/USIM-specific)	140 bytes																			
pid	For Eseye use only. Protocol identifier that indicates the protocol used. The default value is 0 – default store and forward short message.																					
esm	For Eseye use only. User data header that indicates special message attributes associated with <code>short_message</code>. 40 indicates the user data header is included.																					
Replace	Cancel any queued messages																					

Creating 16-bit Unicode messages

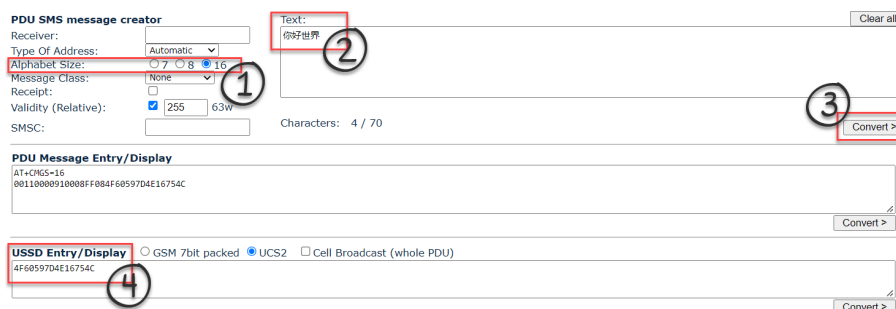
You can convert text into Unicode using freely available internet tools, such as <http://smstools3.kekekasvi.com/topic.php?id=288>.

Example

The Chinese characters for *Hello World* are: 你好世界

1. Using the *SMS Server Tools 3 PDU Converter*, select an *Alphabet Size* of **16**.
2. Type 你好世界 into the *PDU SMS message creator Text* box.
3. Select **Convert >**.

The *USSD Entry/Display* returns the Unicode for 你好世界, which is **4F60597D4E16754C**.



4. Copy and paste the Unicode into your request.

Example

HTTP POST request

<http://messaging.eseye.com/mtsmsapi/index/post>

Example **POST parameters** for this request:

- **username** – UserName
- **password** – UserPassword
- **destination** – 447924939016
- **text (message content)** – Hello World

Example **POST request** using 16-bit Unicode:

```
http://messaging.eseye.com?username=UserName&password=UserPassword&destination=447924939016&message_id=20220302155049&data_coding=8&short_message=4F60597D4E16754C
```

Response text/html

Simple text, either:

- OK!
- **Error<ErrorNumber>: <ErrorDescription>**

For more information, see *Troubleshooting errors* on page 17.

Example

OK!

Requesting a status update for existing SMS messages

Requests a status update for existing AnyNet Messaging Service messages when the following conditions are met:

- The authenticated user has access rights to send an SMS
- If an optional IP address Allowlist exists, the source device is listed on it
- The SIM is not barred from receiving SMS messages



Enabling or disabling SMS Messages is linked to the SIM package, which is outside the scope of the SMS API.

Authentication

- SMS API credentials

Request application/x-www-form-urlencoded

Schema

POST parameters	Description	Required?
username	The Eseye-assigned SMS API username	✓
password	The Eseye-assigned SMS API password	✓
message_id	A user-generated unique message identifier for tracking responses.	✓

Example

HTTP POST request

```
POST http://messaging.eseye.com/mtstatusapi/index/post
```

Example **POST parameters** for this request:

- **username** – UserName
- **password** – UserPassword
- **message_id** – 2554f63f1e5d6ac862fa2b6883edb5b1

Response application/xml

Schema

XML Parameters	Description
statusResponse	The XML object, containing details about the request status.
status	The status is either: blocked – Message blocked by rule sent – Message queued on network error – An error occurred. For more information, see <i>Troubleshooting errors</i> on page 17. queued – Not delivered yet, message awaiting routing delivered – Successful delivery mgmt – Message queued on network
description	Contains a more detailed description of the response.

Example

```
<statusResponse>  
  <status>delivered</status>  
  <description>Message delivered</description>  
</statusResponse>
```

Cancelling undelivered SMS messages

Cancels a sent or queued message when the following conditions are met:

- The authenticated user has access rights to send an SMS
- If an optional IP address Allowlist exists, the source device is listed on it
- The SIM is not barred from receiving SMS messages



Enabling or disabling SMS Messages is linked to the SIM package, which is outside the scope of the SMS API.



If the message is already delivered, the API cannot cancel it.

Authentication

- SMS API credentials

Request application/x-www-form-urlencoded

Schema

POST parameters	Description	Required?
username	The Eseye-assigned SMS API username	✓
password	The Eseye-assigned SMS API password	✓
message_id	A user-generated unique message identifier for tracking responses. The value must match the message_id of the message you want to cancel.	✓

Example

The following example uses the **POST parameters** described in the above table.

HTTP POST request
POST http://messaging.eseye.com/mtcancelapi/index/post

Example POST parameters for this request:

- **username** – UserName
- **password** – UserPassword
- **message_id** – 2554f63f1e5d6ac862fa2b6883edb5b1

Response application/xml

Schema

Parameters	Description
status	The status is either: blocked – Message blocked by rule sent – Message queued on network error – An error occurred. For more information, see <i>Troubleshooting errors</i> on page 17. queued – Not delivered yet, message awaiting routing delivered – Successful delivery
message_id	The returned message identifier, for tracking responses.

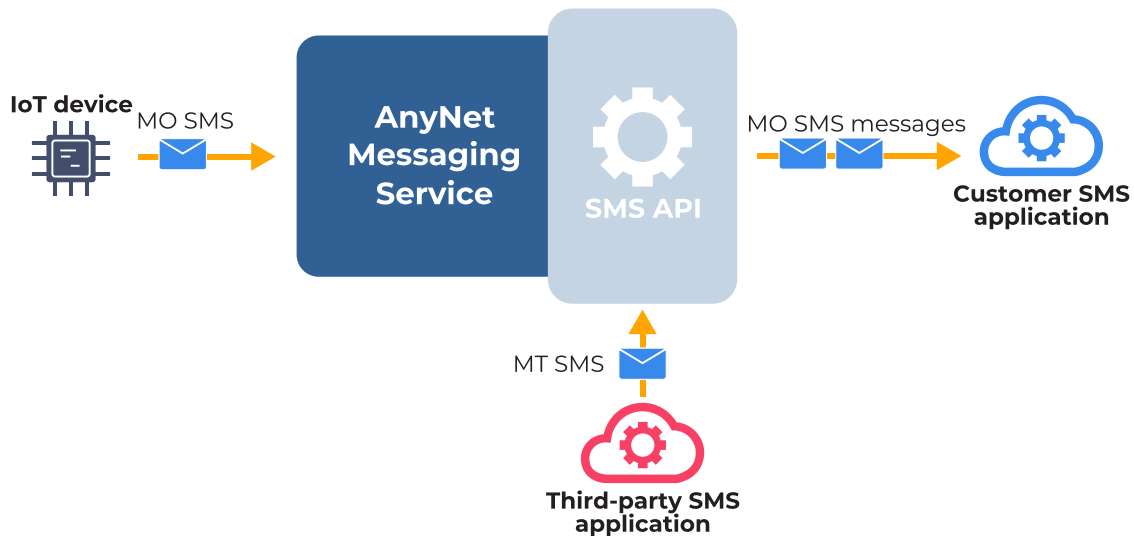
Example

```
<statusResponse>  
  <status>delivered</status>  
  <description>Message delivered</description>  
</statusResponse>
```

Receiving mobile originated SMS messages

Eseye's AnyNet Messaging Service and SMS API enable customers to intercept SMS messages sent to an MSISDN and deliver them to a customer's SMS application.

When IoT devices and other SMS applications send SMS messages to the designated MSISDN ([API MSISDN](#)) linked to the customer's Infinity Classic account, the AnyNet Messaging Service intercepts the SMS messages and delivers them to the customer's application.



To receive mobile originated SMS messages from the SMS API, customers must first:

1. [Set up a HTTP endpoint application](#) – this application receives SMS messages from the SMS API.
2. [Provide details to Eseye Support](#) – these details are typically provided when ordering the SMS API and include the URL of the HTTP endpoint to which the AnyNet Messaging Service will forward messages.
3. [Enable reception of SMS messages in Infinity Classic](#) – using the Infinity Classic web interface.
4. [Send a test MO SMS message](#) – checks that the system is working as intended.

Set up a custom HTTP endpoint application

To receive MO SMS messages, the customer must create an application (HTTP endpoint) that is capable of receiving HTTP POST messages from the SMS API. The SMS message contents (text or data) from the device are contained in the message payload.

MO message format

The AnyNet Messaging Service sends MO SMS PDU messages that may include the fields listed in the table below.



The specific fields depend on the type of PDU (Deliver-SM, Submit-SM, Status-Report, etc) and the information supplied by the originating network.

Field	Type	Description
source	string	The source MSISDN of the originator of the message.  If omitted, the source was an SMS API.
destination	string	The destination MSISDN, which is link to the customer's HTTP SMS endpoint.
message_id	string	A unique identifier for the message that can be used for tracking.
text	string	The human-readable SMS message content.
binary	hexadecimal string	Hexadecimal representation of the Submit-SM PDU.  To decode the contents, use an SMS decoder. Click here for an online SMS PDU decoder.

Provide details to Eseye Support

Once the endpoint application is created, customers must send the following information to Eseye in order to configure their MO SMS application.

- The URL of the HTTP endpoint application , which is stored as the [MO destination URL](#) in the Infinity Classic web interface.
- The Infinity Classic account to which the application is to be linked.



This will determine the charges (if any) for the messages from the SMS API to the customer's application.

Enable reception of MO SMS messages

The AnyNet Messaging Service collects mobile originated (MO) SMS messages and sends them to the customer server when the following conditions are met:

- SMS MO is enabled on the customer's SMS API configuration ([MO Enable option in Infinity Classic](#))
- The source device MSISDN is included on the [SMS allowed list](#)
- If an optional [IP address allowed list](#) exists, the source device is included

To enable reception of MO SMS messages:

1. Log in to the Infinity Classic web interface using the customer account to which the SMS API is linked.
2. Display the Settings > [Messaging API page](#).
3. Under **Actions**, select .
4. Select the **Mobile Originated Enable** check box.
5. If required, edit the **Mobile Originated POST URL**, which is the URL of the customers HTTP endpoint application.
6. Select **Save**.

Test MO SMS message reception

Once configured and enabled, the customer can send an SMS message from an IoT device (or application) to the MSISDN ([API MSISDN](#)) linked to their Infinity Classic account. The AnyNet Messaging Service will intercept the message, look up the MSISDN and re-route it to the customer's application.

Delivery receipts for MT and MO SMS messages

If required, SMS API can automatically send delivery receipts (DR) to your server in response to both MT SMS and MO SMS messages. The original MT SMS and MO SMS messages must include a valid `<receiptUrl>` before the SMS API can post a DR.

The following conditions apply:

- If required, MO SMS is enabled on the customer's SMS API configuration
- The source device MSISDN is included on the SMS Allowlist
- If an optional IP address Allowlist exists, the source device is included

Authentication

- None

Response text/html

Parameters	Description
messageid	A user-generated unique message identifier for tracking the original message
referenceid	Eseye internally generated reference
statuscode	Delivery status code
status	Either: • delivered – The original message was delivered successfully • failed – No delivery confirmation was received

Example

```
http://ota.yourcompany.net/callback.php?messageid=9946e3025608c7106944658dfb1318a8&referenceid=3fd515e674835bb233538f9d218a31d9&statuscode=delivered&status=delivered
```

Troubleshooting errors

When an error occurs, an error code and message are generated in the returned result, in XML format:

```
<statusResponse>
  <status>error</description>
  <description>An error occurred</description>
</statusResponse>
```

The following table can help you diagnose the error:

Error Code and Message	Reason for error
ERROR 502 – Index is not currently a supported method of this API.	Index request not supported.
ERROR 503 – PUT is not currently a supported method of this API.	PUT, GET and INDEX requests are not supported for the SMS API.
ERROR 510 – API Requests from this IP Address (<IPAddress>) are not allowed. where <IPAddress> is the denied IP address	The device used does not exist on the Server IP Allowlist. For more information, see <i>Before you begin</i> on page 1.
ERROR 511 – <Parameter> is required.	Missing mandatory parameter.
ERROR 519 – Unknown parameter: <Parameter>	Unknown parameter supplied.
ERROR 520 – Mode '<value>' is not valid where <value> is the supplied value	The mode parameter is empty, or contains one or more invalid mode specifiers.
ERROR 521 – Mode not specified	The mode parameter is empty, or contains one or more invalid mode specifiers.
ERROR 525 – Max message length of X chars exceeded where X is the limit for the specified mode	The SMS message is too long. Messages are limited to: 160 characters for 7-bit delivery 140 characters for 8-bit delivery
ERROR 526 – message_id must be supplied for delivery receipt	Only applies when a delivery receipt is requested. The message_id is missing.
ERROR 530 – API Requests authentication failed.	The Username or Password value is incorrect.
ERROR 533 – Invalid receipt_url value supplied	The delivery receipt URL supplied is incorrect, invalid, or missing.

Error Code and Message	Reason for error
ERROR 535 – Invalid message_id value supplied	The message_id value is missing.
ERROR 536 – Invalid text value supplied	The Text value is missing.
ERROR 555 – Invalid MSISDN (<MSISDN>) supplied where <MSISDN> is the invalid Mobile Station International ISDN Number (MSISDN)	The supplied MSISDN is incorrect. Ensure you include the appropriate country calling code.
ERROR 555 – Invalid Input Provided	The Query Parameters are incorrect, missing, or invalid.
*ERROR 630 – Invalid MSISDN (<MSISDN>) supplied where <MSISDN> is the invalid Mobile Station International ISDN Number (MSISDN)	The supplied MSISDN is incorrect. Ensure you include the appropriate country calling code.
ERROR 570 –Destination MSISDN (<MSISDN>) has no credit where <MSISDN> is the destination MSISDN	Only applies where the destination MSISDN account is charged. You must add credit to the destination MSISDN before you can use the SMS API with that MSISDN. For information about adding credit to your account, see Making a payment .
ERROR 580 – Please use only GSM 03.38 basic 7 bit characters, or specify 8 bit mode	A non-standard character exists in the message text. For 7-bit mode, SMS API only supports characters from the ETSI USSD default alphabet, as defined in GSM 03.38 .

Coding non-alphanumeric ASCII characters

Character	Description	Encode as
[Carriage Return]	carriage return	%0D
[Line feed]	line feed	%0A
[Space]	space	%20
"	quotation mark	%22
#	hash	%23
%	percent	%25
&	ampersand	%26
,	comma	%2C
.	period	%2E
/	forward slash	%2F
;	semi-colon	%3A
:	colon	%3B
<	less than	%3C
=	equal	%3D
>	greater than	%3E
?	question mark	%3F
!	exclamation mark	%A1
£	pound	%A3
¤	currency sign	%A4
¥	yen	%A5
§	paragraph sign	%A7
Ä	capital A with diaeresis	%C4
Å	capital A with ring	%C5
à	small a grave	%E0
ä	small a with diaeresis	%E4
å	small a with ring	%E5

Character	Description	Encode as
Æ	capital diphthong AE	%C6
Ç	capital C cedilla	%C7
É	capital E acute	%C9
è	small e grave	%E8
é	small e acute	%E9
ì	small i grave	%EC
Ñ	capital N with tilde	%D1
ñ	small n with tilde	%F1
ò	small o grave	%F2
ö	small o with diaeresis	%F6
Ø	capital O with storke	%D8
Ö	capital O with diaeresis	%D6
Ü	capital U with diaeresis	%DC
ù	small u grave	%F9
ü	small u with diaeresis	%FC
ß	small s sharp	%DF



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